Digital Avionics
航電系統

Lecture 2

Prof. Ying-Chih Lai

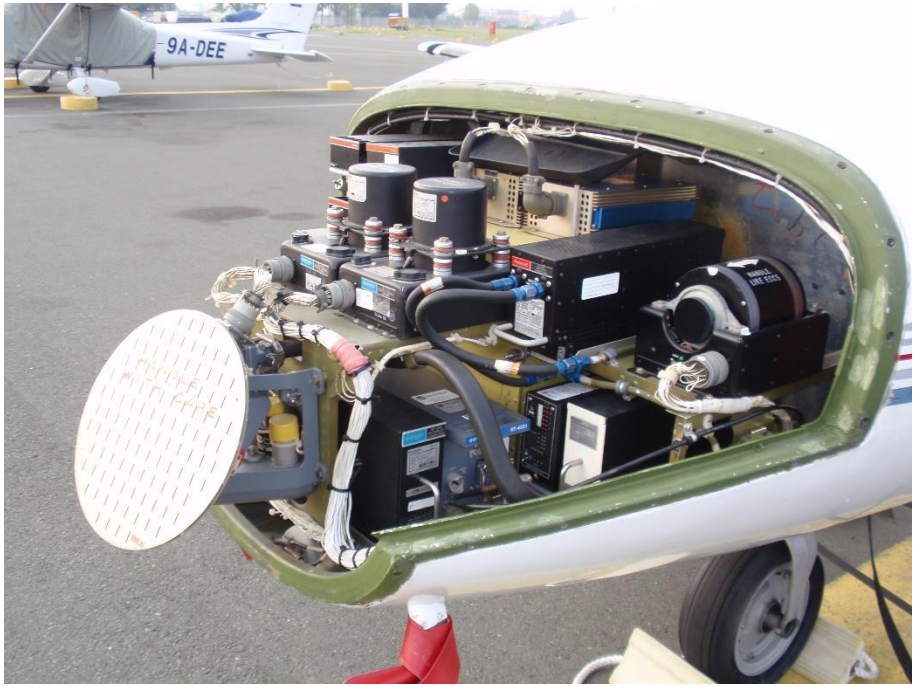
Textbook

- Introduction to Avionics Systems, 3rd Edition.
- Reading:
  - Chapter 1, pp. 1-17

Avionics

- 'Avionics' is a word derived from the combination of aviation and electronics. It was first used in the USA in the early 1950s and has since gained wide scale usage and acceptance although it must be said that it may still be necessary to explain what it means to the lay person on occasions.
- The avionics equipment on a modern military or civil aircraft can account for around 30% of the total cost of the aircraft. This figure for the avionics content is more like 40% in the case of a maritime patrol/anti-submarine aircraft (or helicopter) and can be over 75% of the total cost in the case of an airborne early warning aircraft (AWACS).

Avionics



Radar and other avionics in the nose of a Cessna Citation I/SP



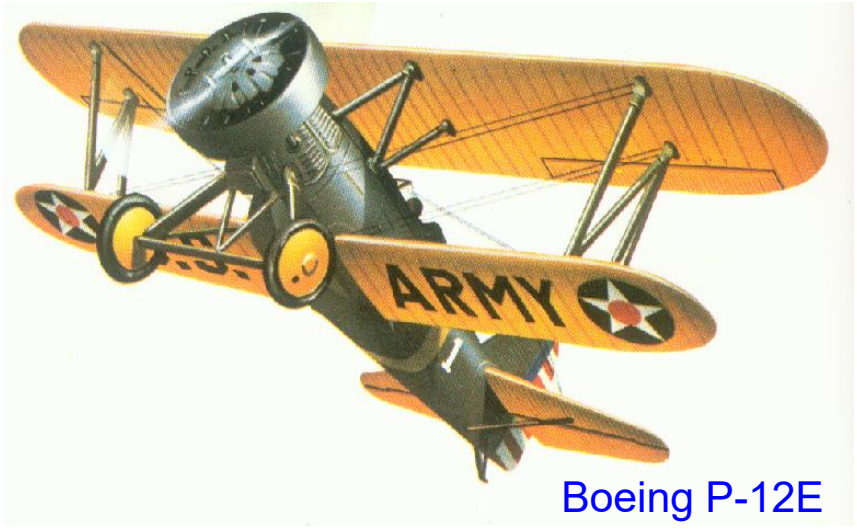
Cockpit Voice Recorder (CVR) – AV32
Flight Data Recorder (FDR)
Nav/Com – AV34
GPS/Nav/Com – AV35
Transponder – AV36
Flight Director – AV37
Horizontal Situation Indicator (HSI) – AV38
Audio Panel – AV39

Age of Feel Flight

- Before 1920, pilot depended on visual capability and personal feeling to fly an aircraft, just like ultra-light flight today.
- Hear wind speed, estimate ground altitude, see horizontal line, search for destination by ground feature. But not of night flights.
- Speed<100 knots, altitude<3000 feet, range<200 miles



France S.7



Boeing P-12E

Units

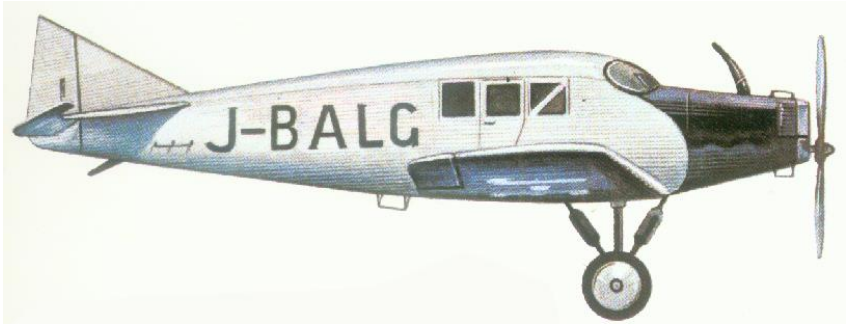
- The knot is a unit of speed equal to one nautical mile per hour, exactly 1.852 km/h (approximately 1.15078 mph).



1 kn in ...	*... is equal to ...*
km/h	1.852
mph	1.15078
m/s	0.514444
ft/s	1.68781

Instruments for Visual Flight

- From 1920～1940, pilots expected visual all weather flights for low visibility, night raining and foggy season.
- Aircraft needed to develop navigation, communication and instruments beyond speed, altitude and horizon.
- Speed<150 knots, altitude < 5000 feet, range < 600 miles



German F-13



Boeing 314

Early Avionics Information

- Early avionics system assisted pilot to fly visual flight rule (VFR) for speed, altitude and horizon.
- In advance to instrument flight rule (IFR) for navigation
- Fulfilled basic demand for night flights



UAS B-24

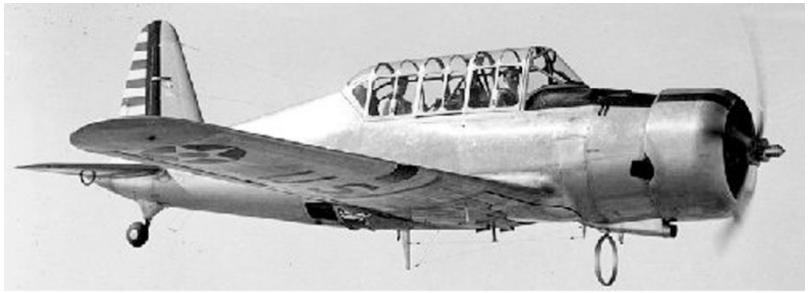


Instrument panel of the B-24J



Early Basic Avionics

- Basic instruments offers messages for wind speed, altitude
- Simple gyro for attitude and horizon
- Baro-height meter for Mean Sea Level (MSL)
- Audio low frequency radio tone as navigation signal



UAS BT-13



Instrument Panel

Avionics System Coverage

- Before WWII of propeller era, aircraft system mainly contained thrust power and operation mechanics
- Modern avionics contains widely to all related electronics systems, subsystems, modules, elements, sensors in integration.

Stand Alone Avionics

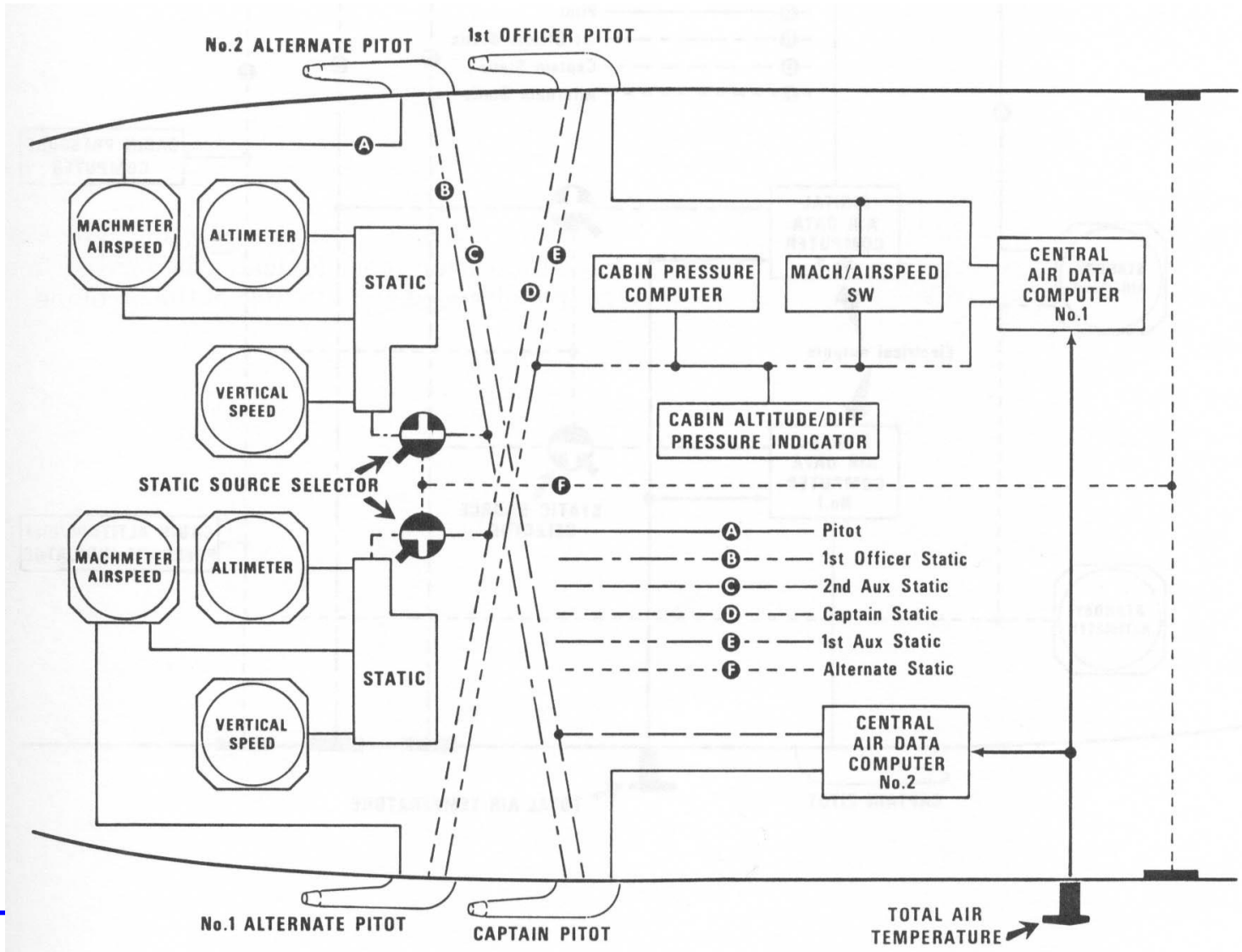
- Early avionics also called flight instruments as Stand Alone to each other
- One instrument usually coped with its sensor or transducer, multiple elements were used.
- Pilots needed good vision sensitivity to read all instantaneously message changes from all different instruments.



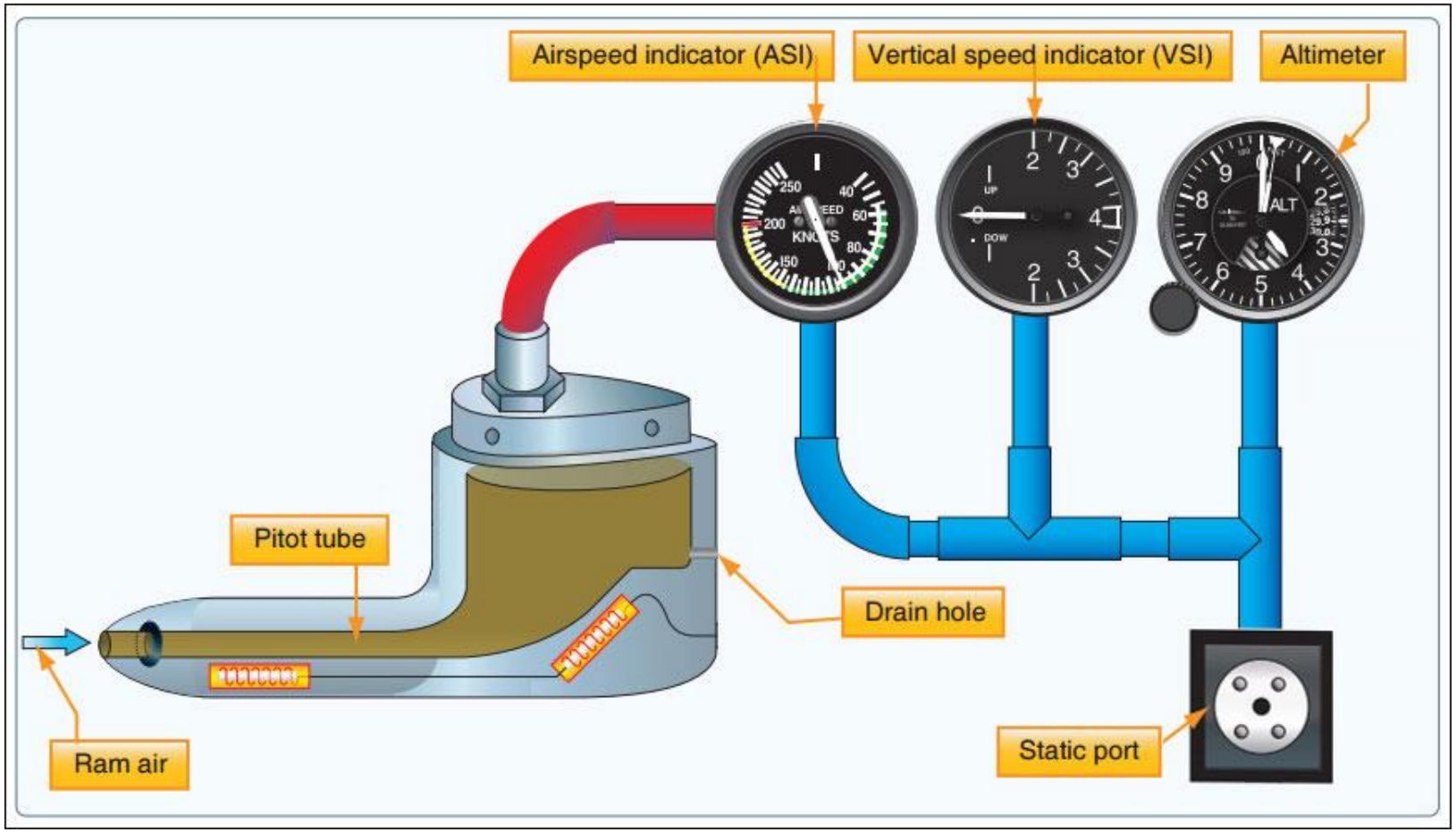
Stand Alone Avionics Structure

- Analog sensor and analog transmission
- Individual circuit to transmit signals
- Sensors are multiple to different meters
- Analog signal error from circuit path
- Significant electromagnetic Interference (EMI), electro-static and noise from environment

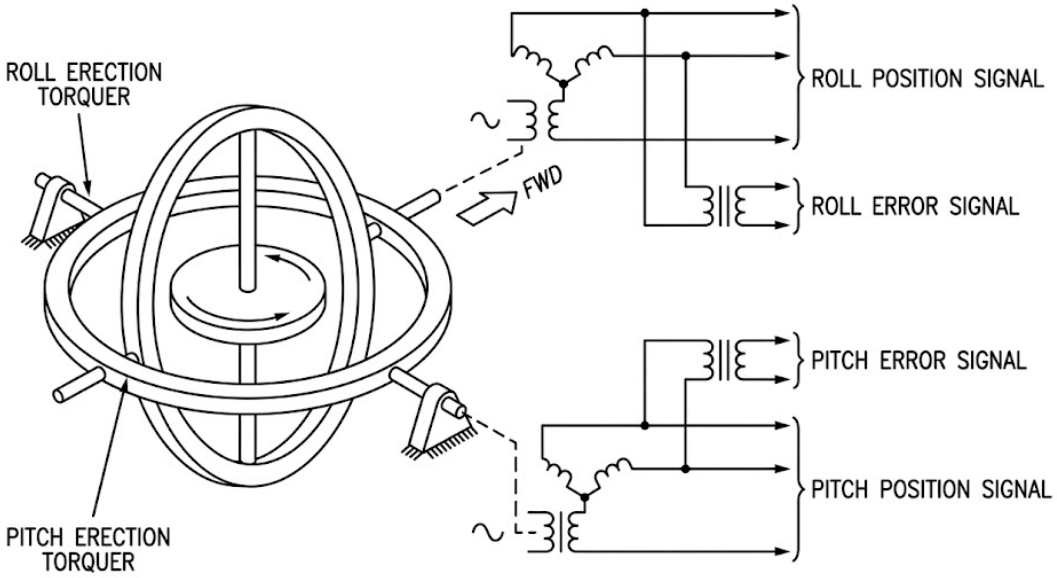
B-757 Avionics

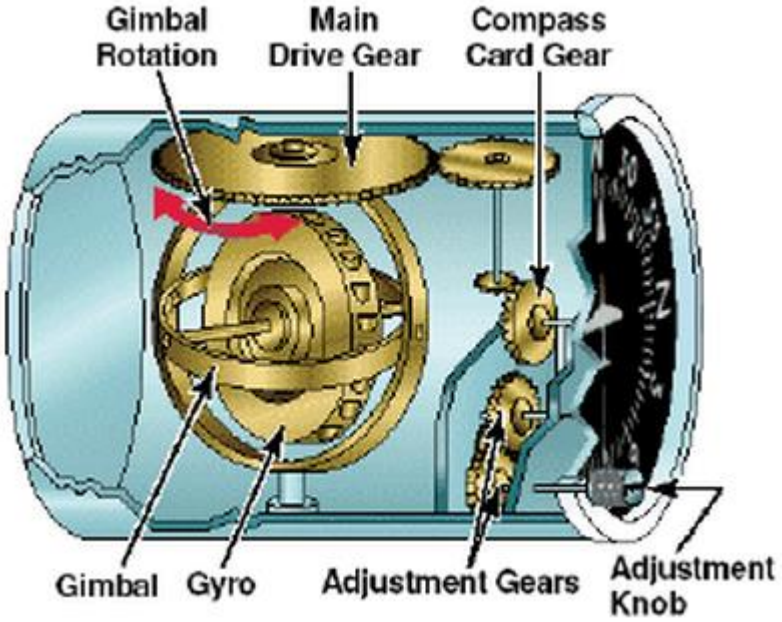


Stand Alone Instrument

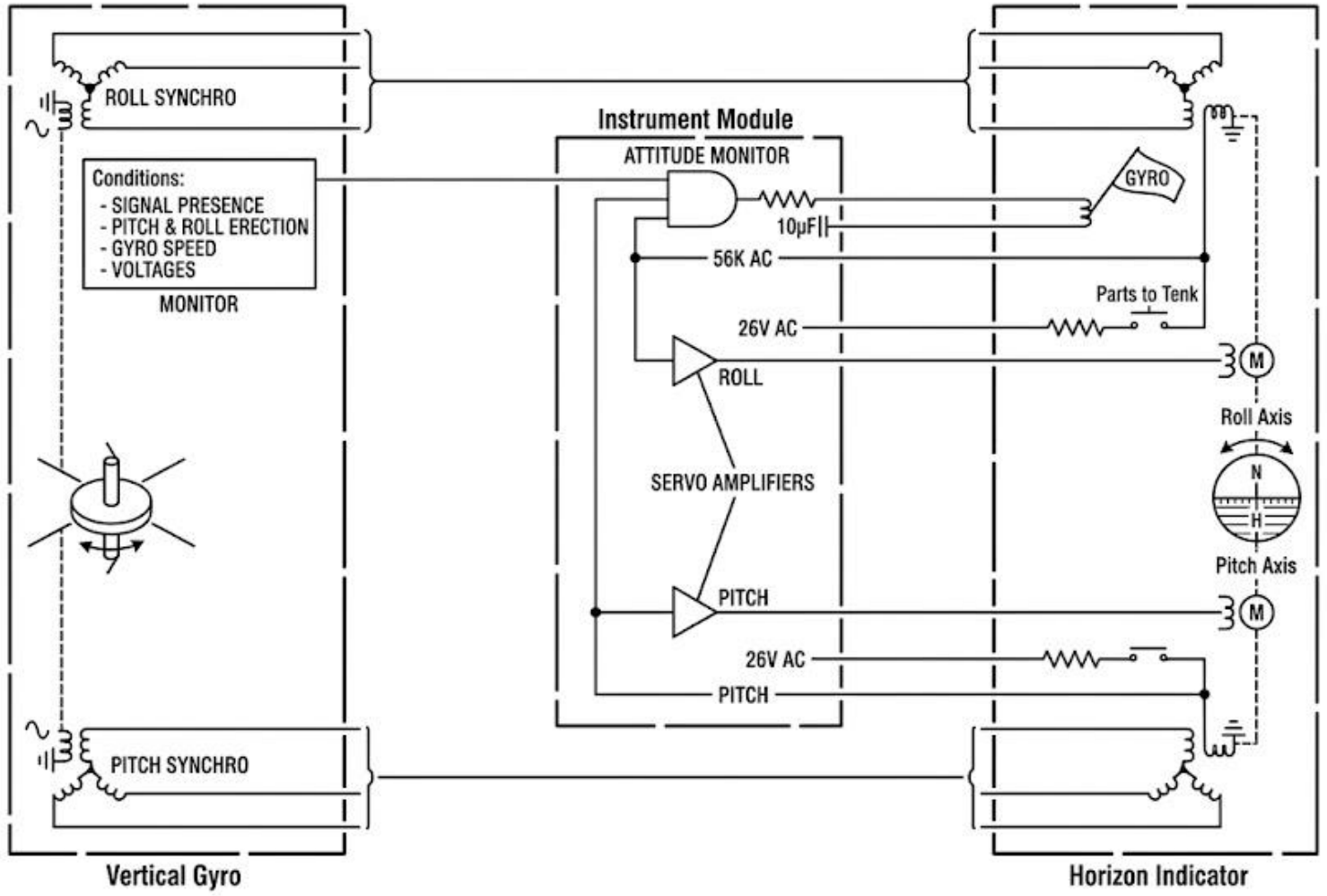


Stand Alone Instrument





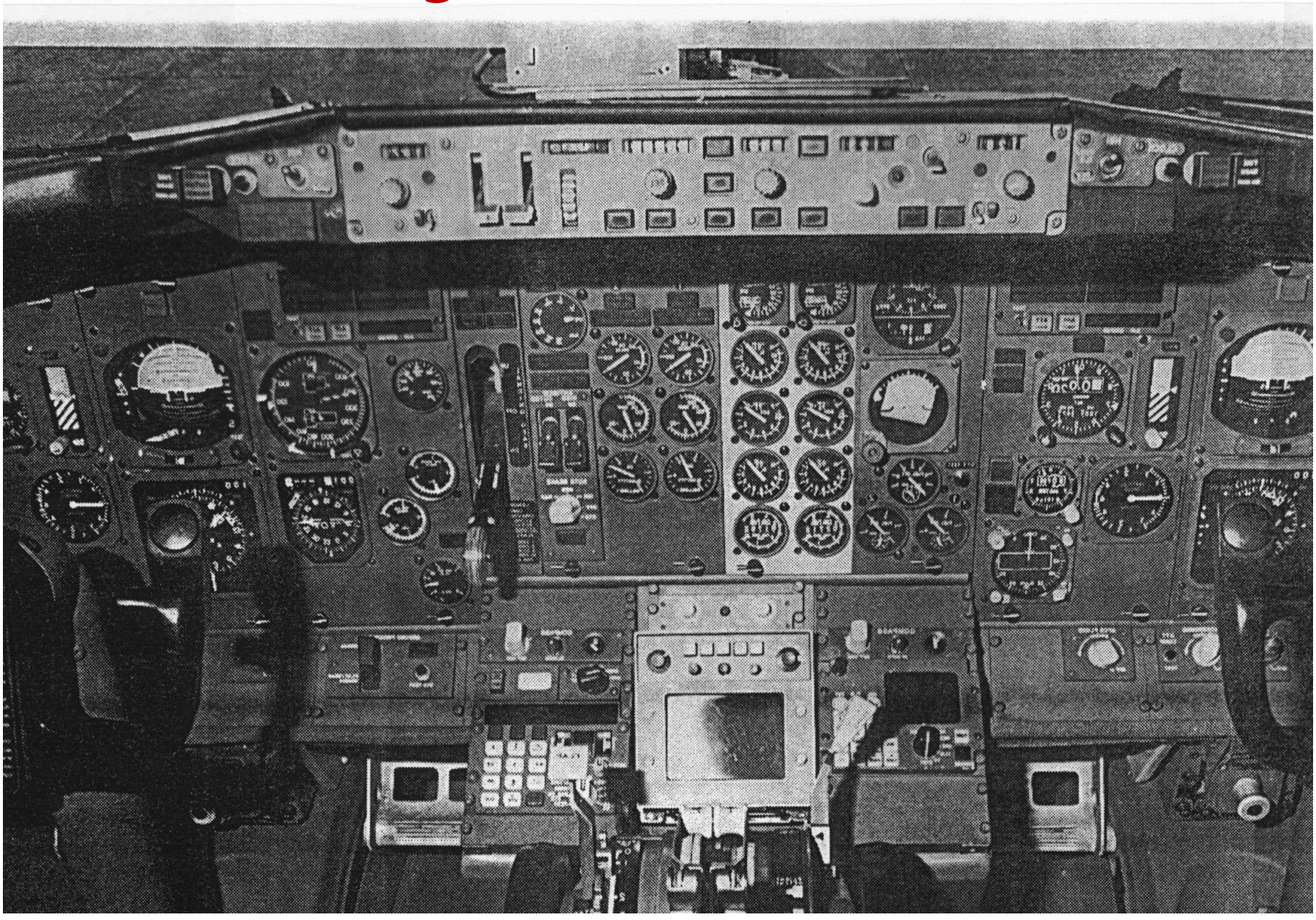
Stand Alone Analog Instrument



Stand Alone Instrument

- Use analog Synchro to transmit analog signal by 1:1 delivery within 3 meters
- One meter coped with one sensor source
- Each instrument is independent, cannot integrate.
- Airborne computer reads data from different sensor sources

B-737 Analog Avionics



B-737 Digital Avionics



Early Cabin Crew

- Early cabin crew included 4 pilots
- Captain and first officer (F/O)
  - As pilot flight (PF) or pilot not flight (PNF)
- Two second officers (S/O) for navigation and communication
  - Supporting navigation and communication



747-200 cockpit

Digital Avionics System

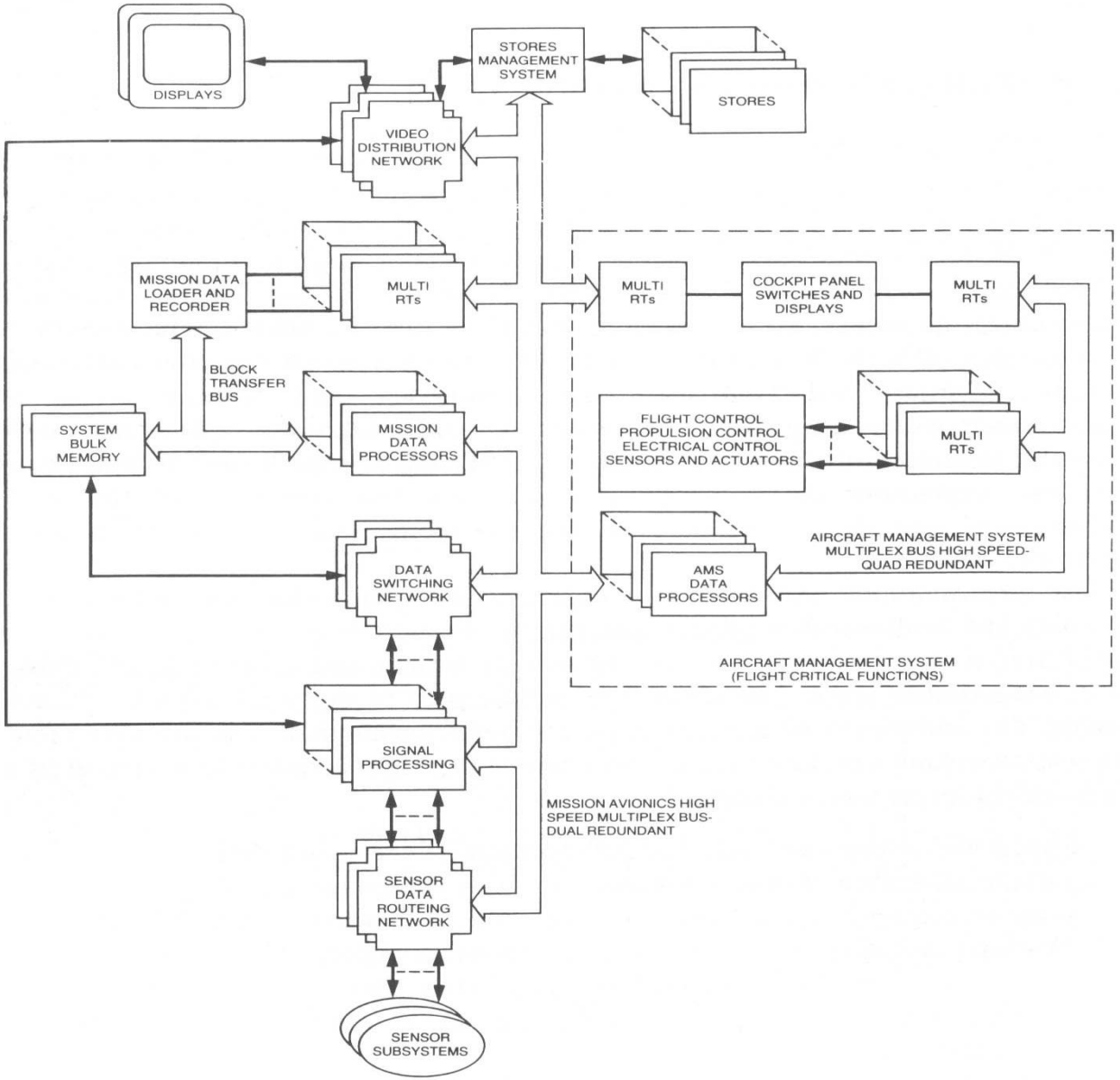
- Under digital system configuration, all information are linked by data bus to integrate into one operation network and share by controller and displays.
- After integration, multiple messages are presenting to pilot on different phases of flight operation.



Digital Avionics Structure

- Simplify sensor numbers to specific location
- All messages are converted into digital current pulse trains
- Use digital data bus to input and output messages
- Each unit is assigned with an address number, controlled by different pulse clock to distinguish control area and control speed

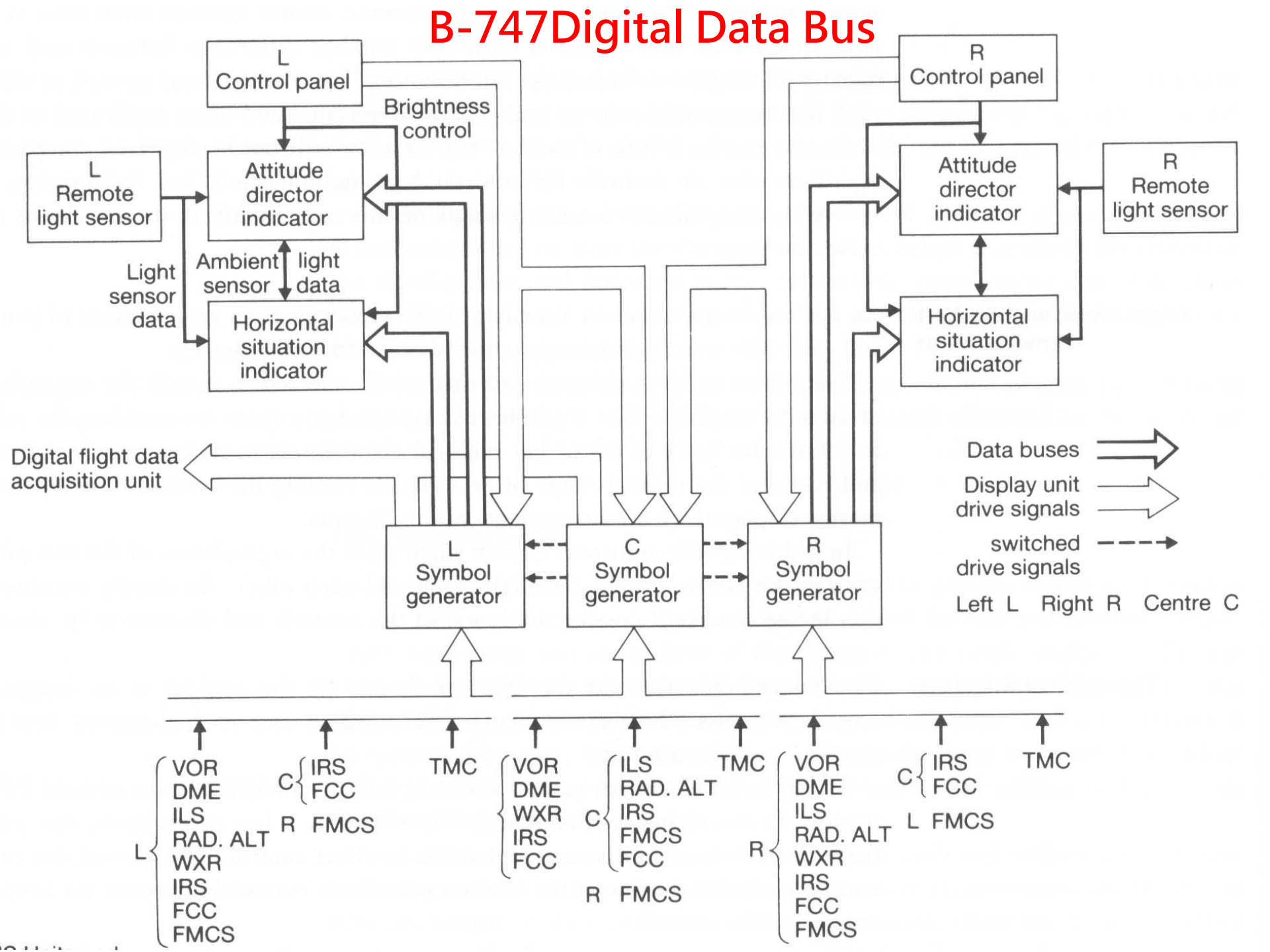
Digital Data Bus



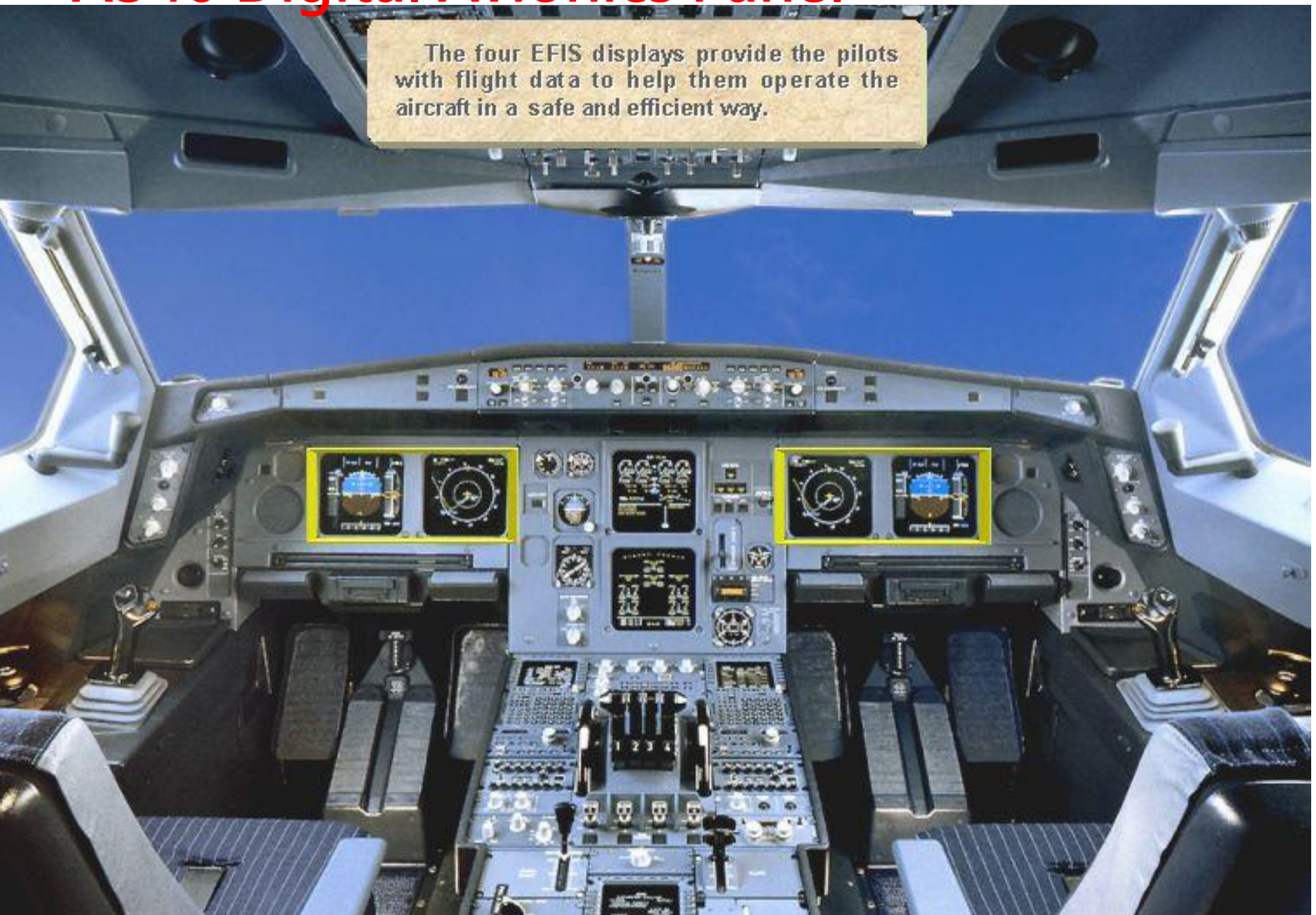
Digital Data Bus Control

- Digitalized avionics system establish data bus to exchange bidirectional message flows from remote terminals to hierarchical computers for automatic control
- Central computer and local computer carry different CPU rate of control commands, such as high speed in flight stability and low speed in autopilot navigation

B-747Digital Data Bus



A340 Digital Avionics Panel



Cabin Crew in Digital Avionics

- After 1980, B-747 flight crew has been reduced into captain and first office only for PF and PNF to share flight, navigation, communication and maneuvering
- After 1985, cabin crew changed again due to long haul capability (10 hour limit for each pilot), more pilots to share flight hours

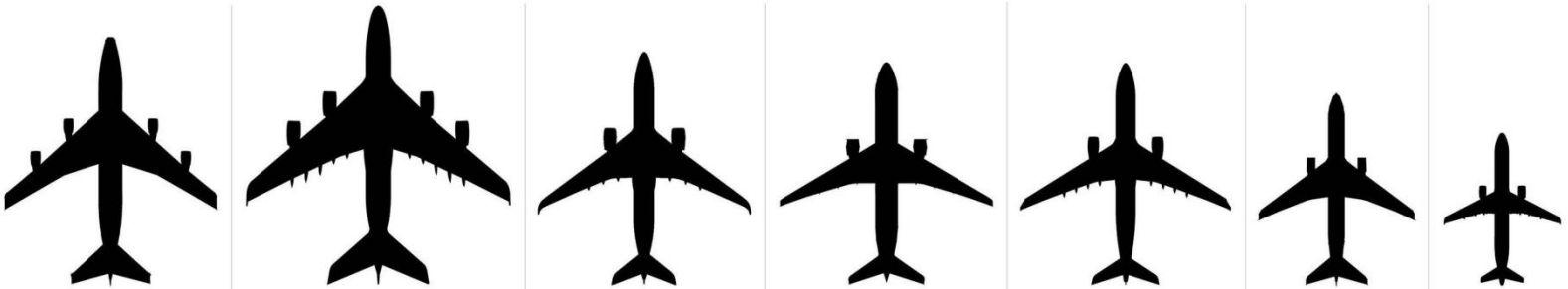
Cabin Crew in Digital Avionics

- Flight duty period limitations of flight crew:

1. Single flight crew. An operator may assign a pilot to a flight duty period up to 12 consecutive hours for domestic flights or a flight duty period up to 14 consecutive hours for international flights. If the domestic and international flights are dispatched during single assignment, duty time limitation should follow that of international flights.
2. Multiple flight crew. An operator may assign a pilot to a flight duty period up to 18 consecutive hours.
3. Double flight crew. An operator may assign a pilot to a flight duty period up to 24 consecutive hours.

Aircraft System Enhancement

- Enhancement to aircraft system performance
- Increase longer flight range
- Increase cruise speed
- Increase payload



BOEING 747-400
Wingspan: 213 feet
Height: 62 feet
Length: 232 feet
Cruising speed: 564 mph
Range: 7,700 miles*
Passengers: 376*

AIRBUS A380
(does not fly through Minneapolis)
Wingspan: 262 feet
Height: 79 feet
Length: 239 feet
Cruising speed: 561 mph
Range: 9,436 miles
Passengers: Typically 544, up to 853

AIRBUS A350-900
(replacing Delta's 747s, but may not fly through Minneapolis)
Wingspan: 212 feet
Height: 56 feet
Length: 219 feet
Cruising speed: 561 mph
Range: 9,321 miles
Passengers: 325

BOEING 777-200ER
(extended range)
Wingspan: 212 feet
Height: 61 feet
Length: 209 feet
Cruising speed: 550 mph
Range: 8,542 miles*
Passengers: 291*

AIRBUS A330-300
Wingspan: 198 feet
Height: 56 feet
Length: 209 feet
Cruising speed: 531 mph
Range: 5,343 miles* (extended to 8,400 miles with the similarly sized A340-300, which has two more engines)
Passengers: 293*

BOEING 767-300ER
(extended range)
Wingspan: 156 feet
Height: 52 feet
Length: 180 feet
Cruising speed: 517 mph
Range: 6,221 miles*
Passengers: 226*

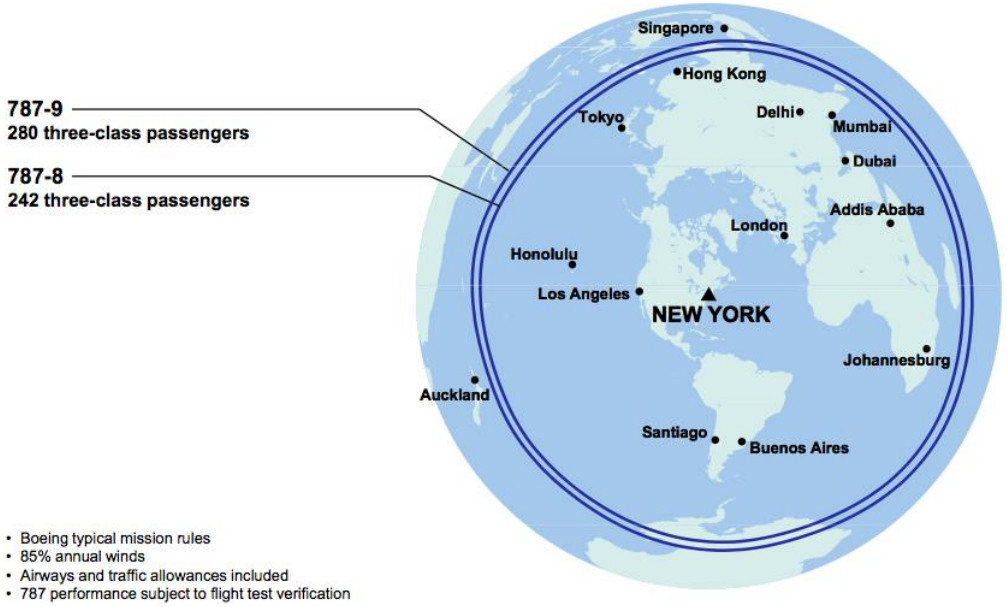
AIRBUS A321
Wingspan: 117 feet
Height: 38 feet
Length: 143 feet
Cruising speed: 517 mph
Range: 2,565 miles*
Passengers: 192*

*Delta configuration

SOURCES: DELTA AIR LINES INC., BOEING CO. AND AIRBUS

Aircraft System Enhancement

- Improve navigation and communication
- Improve engine propulsion thrust
- Enhance system operation function and capability
- Elevate aircraft availability and maintainability



Aviation Developments

- Before 1937
- Cruise speed < 150 n miles, range< 500 miles, on altitude 5000 feet
- Propeller powered
- Dominantly VFR for small capacity passenger flights, post, agriculture, farm and forest applications

Avionics Developments

- Before 1980, mainly stand alone instruments
- Speed <500 n miles, range 5000 miles, altitude 40,000 feet
- Jet propulsion with propeller powered
- Dominantly IFR and partly VFR
- Mass transportation on passenger and cargo, such as B-747

Modern Avionics

- After 1980 till now, digital avionics system has systemized integrated into automation
- Computer operation replaces pilot manipulation
- Integral instrument displays multiple information at one monitor
- Airway and navigation are integrated into avionics system.



Modern CNS Avionics

- In 1995, GPS has been approved into avionics for navigation
- Satellite systems are widely applied into communication, navigation and surveillance with fully system automation
- Digital avionics reduces weight and size in a great deal, digital data bus exchanges mass message in high speed in wired and wireless

Broad Avionics Categories

- Broad avionics system contains airborne avionics system and ground AT avionics system
- Avionics and ATC are two of a kind, using the same technology for reverse purposes

Broad Avionics Categories

- Ground ATC avionics system follows ATC procedures to monitor and separate aircraft via communication for flight safety
- Airborne avionics system focuses on operation and control of aircraft by communication, navigation, surveillance and automation to a flight mission

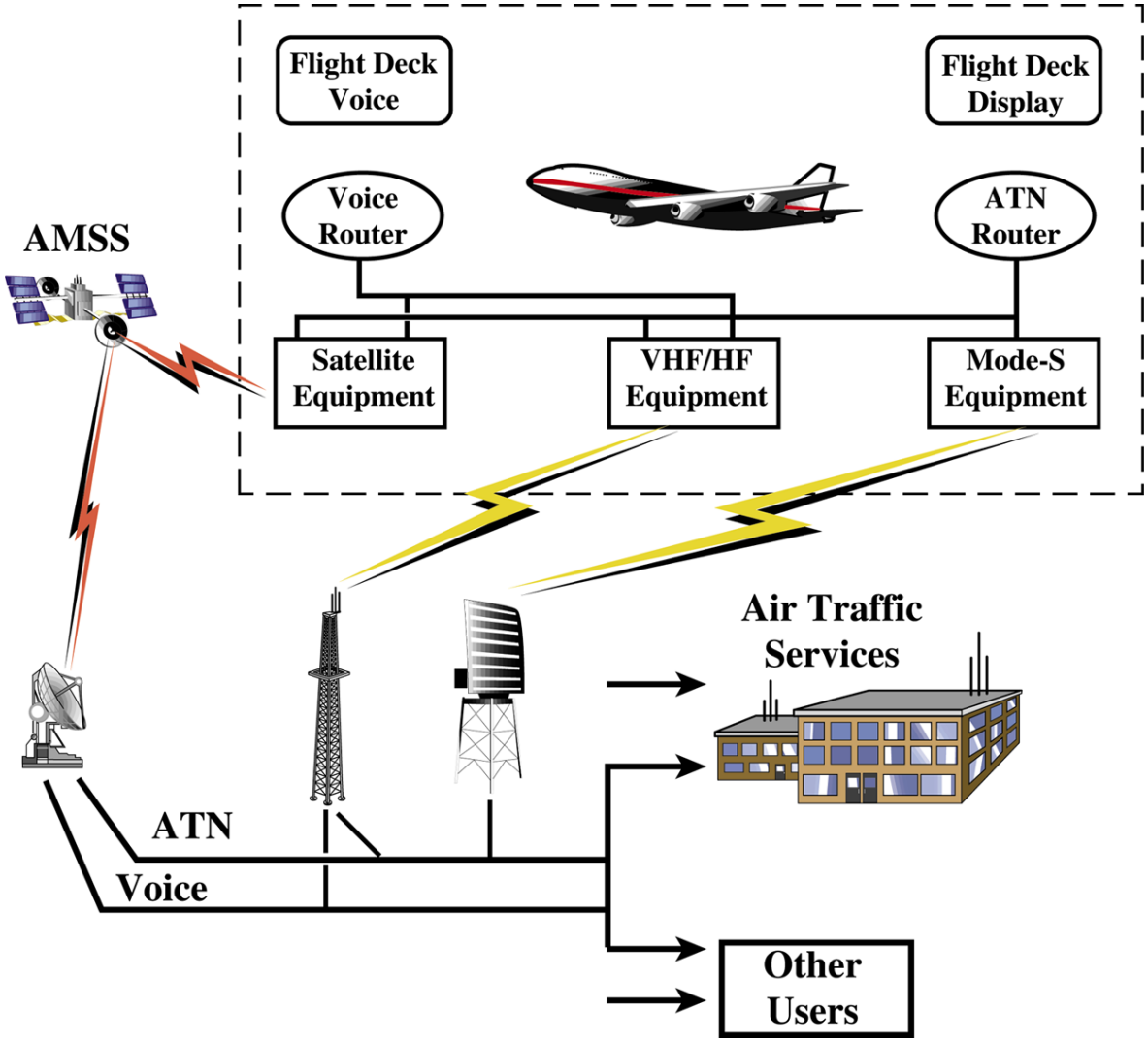
Conventional Avionics CNS

- Conventional CNS includes:
- C—use HF, VHF, UHF for A-G voice communication
- N—use ground facility of VOR, FIX to build airways for flight navigation
- S—use radar for search into airspace and aware of aircraft movements for surveillance

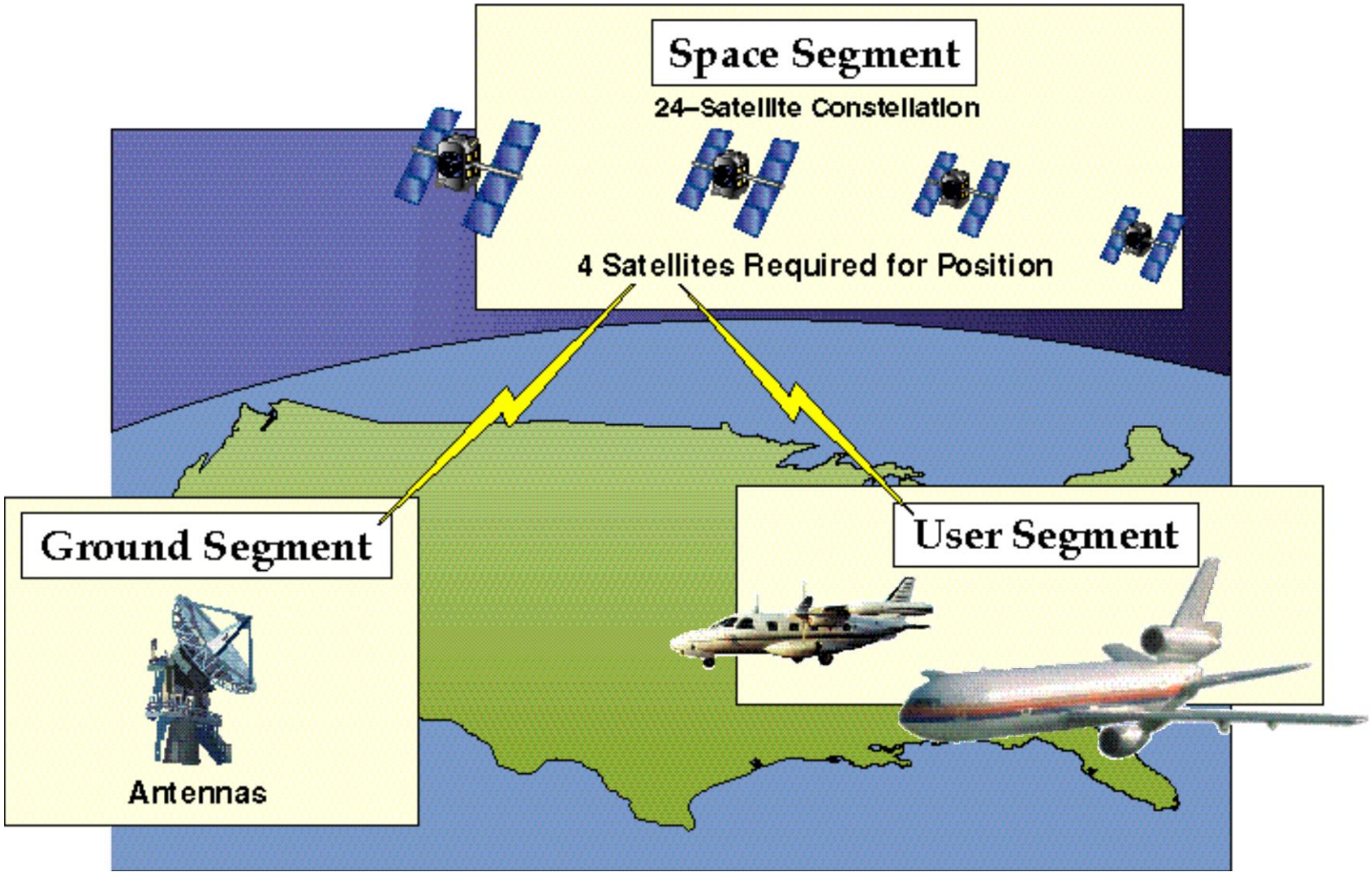
Broad Sense on Avionics CNS

- Future avionics CNS includes:
- C—use HF, VHF, UHF, SATCOM for A-A or A-G voice and data communication
- N—use GNSS to establish WAAS and LAAS for seamless navigation by free flight
- S—use ADS-B to report aircraft dynamic report for ATC surveillance

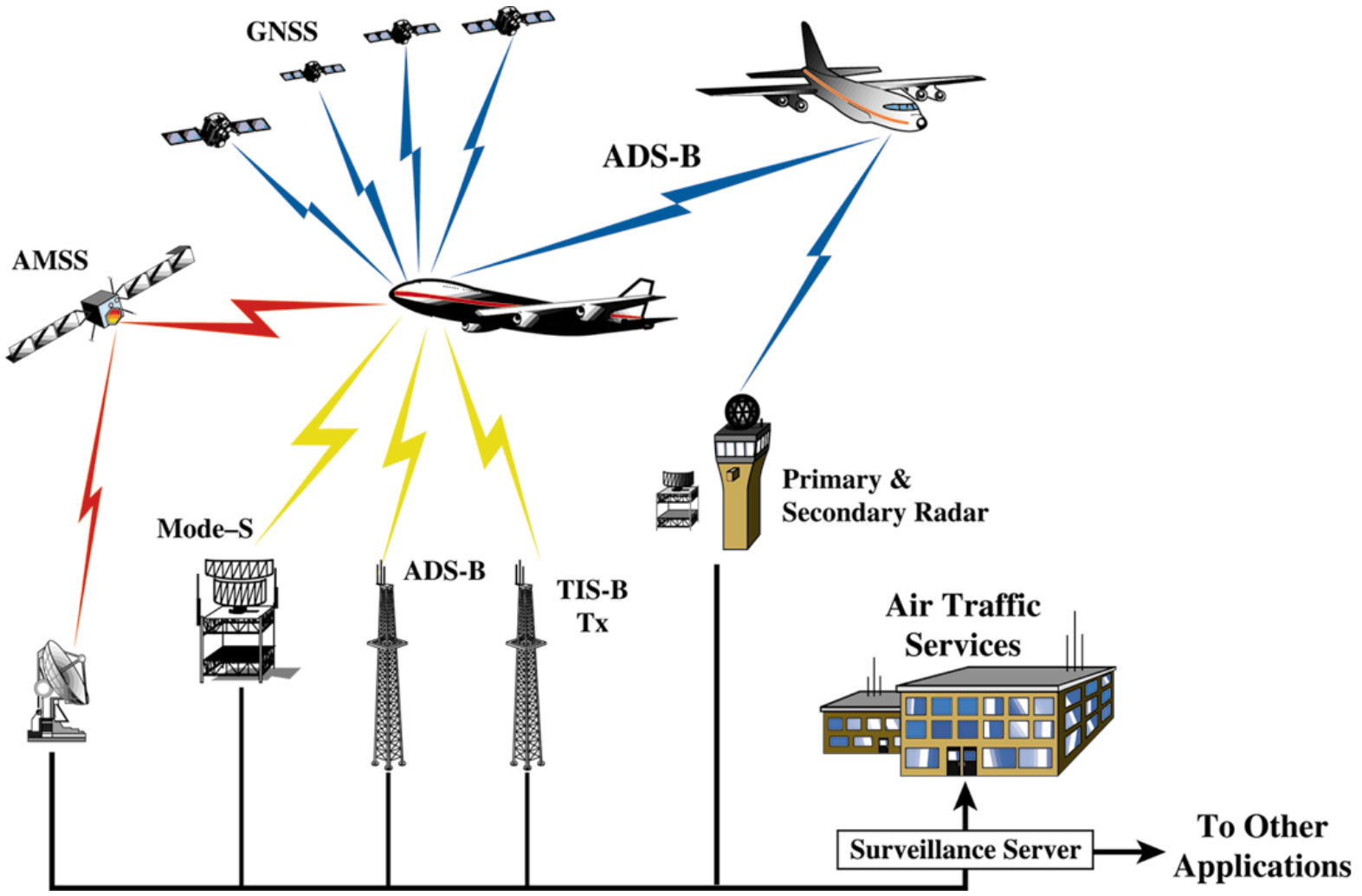
Broad Communication Avionics



Broad Navigation Avionics



Broad Navigation Avionics



Digital Stand Along Meter



ULA Instruments

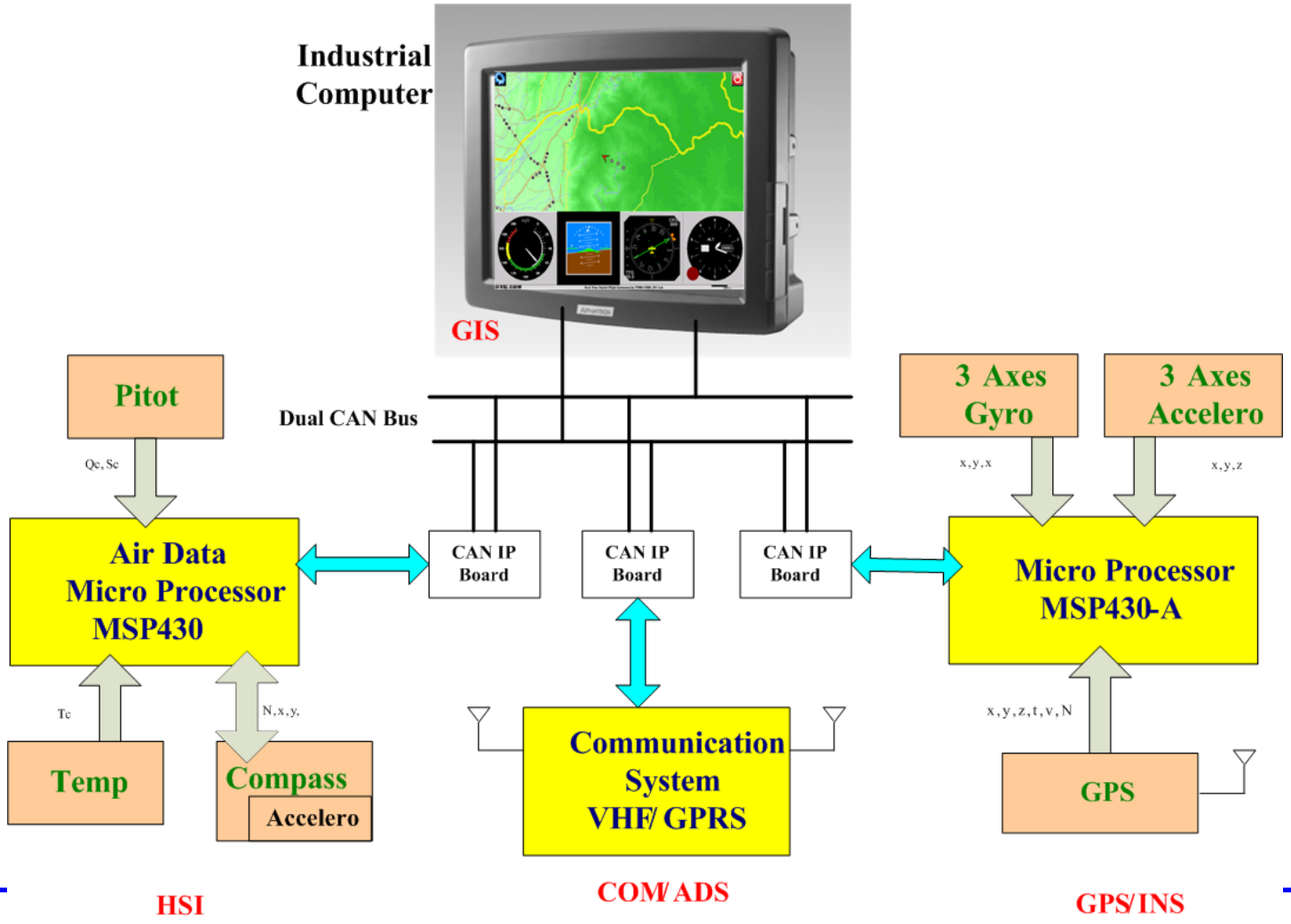




Avionics Design Concept

- Digital avionics adopts Integrated Module Avionics (IMA) to construct circuit modularization
- Each IMA is functionally independent, and is capable as Line Replaceable Unit (LRU)
- Electronics technology of very-large-scale integration (VLSI), Microelectromechanical system (MEMS) and microprocessor is integrated through Data Bus for mass data exchange

ULA Avioncs Panel



GA Avionics Panel-Cessna 172 SP



GA Avionics Panel-Cessna 172 SP



Aircraft Avionics Panel-A320



Aircraft Avionics Panel-A350



Aircraft Avionics Panel-A380



Aircraft Avionics Panel-Boeing 787



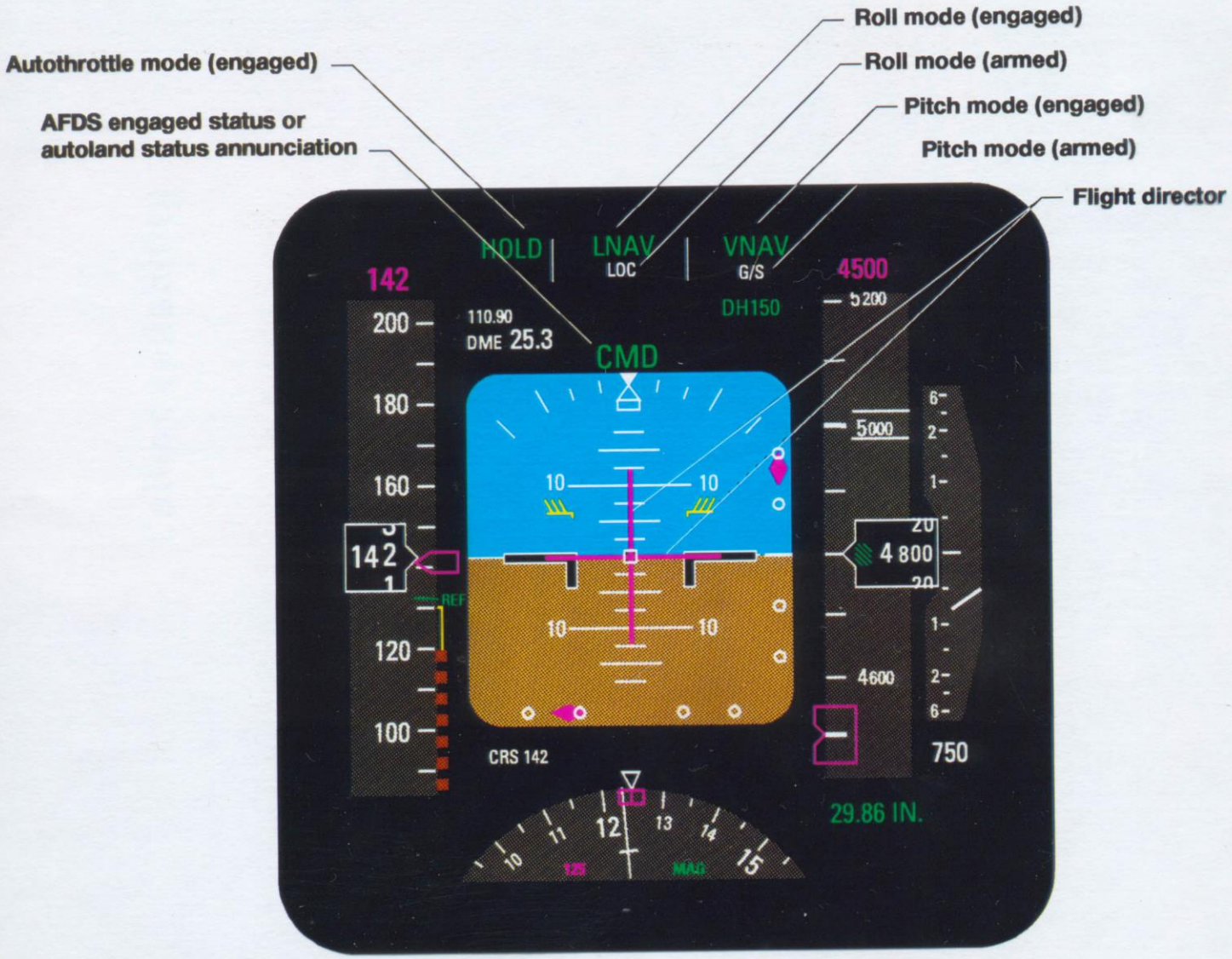
Digital Avionics Instrument Panel

747-400 CRT/LCD Transition



AFDS Mode Annunciation

747-400



Digital Avionics BIT Maintenance



Digital Avionics Maintenance

- Digital avionics maintenance change the past rules to check each subsystem on site
- The Integrated Module Avionics (IMA) embedded on-line Built-in Test (BIT) modules into all subsystem
- Inflight malfunctions will be read and recorded inflight computer.

Digital Avionics Maintenance

- Maintenance engineer download BIT records on aircraft arrival.
- Identify all malfunctions and assess faults
- Replace malfunction parts accordingly from avionics warehouse.
- The driven parts will be delivered to avionics laboratory for examination and refurbishment.

Ground ATC avionics is also enhanced





